Healthcare Patient Analytics

Patient Admissions, Hospital Billing, and Length-of-Stay Analysis

1. Project Overview

This project analyzes a healthcare dataset containing patient demographics, admission details, medical conditions, billing amounts, discharge dates, medications, and test results. The goal is to summarize patient admissions, compare recorded billing across categories, examine length of stay, and identify billing records that require review. The workflow combines Python for cleaning and exploratory analysis, MySQL for structured business questions, and Power BI for interactive reporting.

2. Dataset Summary

Item	Details
Original dataset	55,500 rows × 15 columns
After exact duplicate removal	54,966 rows
Columns after adding review flag	16
Missing values in original columns	0 reported in notebook
Exact duplicate rows removed	534
Age range	13–89 years
Billing range	-$2,008.49 to $52,764.28
Records flagged for billing review	106 (billing amount ≤ 0)

Scope note: The dataset includes patient records and hospital names, but does not provide staff/employee data or a readmission indicator. Therefore, this report does not make claims about employee performance, readmissions, treatment effectiveness, or causes of clinical outcomes.

3. Data Cleaning and Preprocessing in Python

The notebook uses pandas to inspect, clean, standardize, and export the healthcare dataset.

Data loading and inspection

Loaded the CSV into a pandas DataFrame and reviewed its shape, columns, sample rows, data types, and descriptive statistics.

Missing-value check

Checked every original column for null values. The notebook reports zero missing values in the original 15 columns.

Column-name standardization

Converted column names to lowercase snake_case and renamed the admission date field to admission_date.

Date conversion

Converted admission_date and discharge_date to datetime data types for date-based analysis.

Duplicate removal

Identified 534 exact duplicate rows and removed them, reducing the dataset from 55,500 to 54,966 rows.

Text consistency

Trimmed text values and standardized selected categorical fields to improve consistency.

Age validation

Checked missing ages and values outside a screening range. The cleaned data has ages from 13 to 89, with no missing or out-of-range ages reported.

Billing review flag

Reviewed billing statistics and created billing_review_needed for records with billing_amount ≤ 0. The notebook identified 106 negative billing records; they were flagged for review rather than silently removed.

Export and database load

Saved the cleaned dataset and loaded it into the MySQL table patient_data in the healthcare_analytics database.

Data-quality observations

• The original data has no reported null values, so null imputation was not needed for the fields checked.

• The billing distribution includes negative amounts. These may represent data issues or adjustments; the project flags them for investigation and does not assume their meaning.

• Exact duplicate removal is based on all columns matching. It does not establish that two different records for the same person are duplicates.

4. Exploratory Data Analysis (EDA)

EDA was used to understand the cleaned data before asking more specific business questions.

4.1 Numerical summary

Metric	Age	Billing amount
Count	54,966	54,966
Mean	51.54 years	$25,544.31
Median	52 years	$25,542.75
Minimum	13 years	-$2,008.49
Maximum	89 years	$52,764.28
Standard deviation	19.61 years	$14,208.41

4.2 Category distributions

Category	Records
Medical condition — Arthritis	9,218
Medical condition — Diabetes	9,216
Medical condition — Hypertension	9,151
Medical condition — Obesity	9,146
Medical condition — Cancer	9,140
Medical condition — Asthma	9,095
Admission type — Elective	18,473 (33.61%)
Admission type — Urgent	18,391 (33.46%)
Admission type — Emergency	18,102 (32.93%)

The six medical-condition groups and three admission types have broadly similar record counts. Elective admissions are the largest of the three admission categories, but account for about one-third of records rather than a majority. These are descriptive patterns in this dataset, not evidence of cause.

4. Exploratory Analysis — Billing Questions

The notebook grouped billing amounts by medical condition and admission type.

Medical condition	Average recorded billing
Obesity	$25,804.36
Diabetes	$25,660.48
Asthma	$25,633.46
Arthritis	$25,511.78
Hypertension	$25,503.06
Cancer	$25,152.32

Average billing by admission type

Admission type	Average recorded billing
Elective	$25,612.14
Urgent	$25,514.53
Emergency	$25,505.33

Initial interpretation

• Obesity has the highest average recorded billing among the six conditions in the notebook output; Cancer has the lowest.

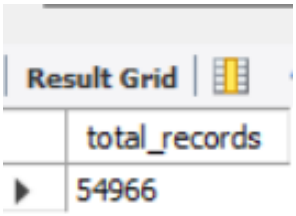
• Elective admissions have the highest average recorded billing among admission types, though the differences are modest.

• Because the dataset is observational and includes negative billing records, these averages should be treated as descriptive and reviewed with billing context before operational decisions.

5. MySQL Analysis — Business Questions

The following questions were answered using the healthcare_analytics database and patient_data table.

Q1. How many patient records are in the database?



The query result shows 54,966 patient records after duplicate removal.

Q2. What are the average, minimum, and maximum ages?



The result shows an average age of 51.5 years, a minimum of 13, and a maximum of 89.

5. MySQL Analysis — Business Questions

The following questions were answered using the healthcare_analytics database and patient_data table.

Q3. How many patient records belong to each gender?

Result Grid | Filter R

gender	total_records
Male	27496
Female	27470

The result shows 27,496 male records and 27,470 female records.

Q4. How many records are there for each medical condition?

Result Grid | Filter Rows:

medical_condition	total_records
Cancer	9140
Obesity	9146
Diabetes	9216
Asthma	9095
Hypertension	9151
Arthritis	9218

Arthritis has 9,218 records and Diabetes has 9,216; the six conditions have similar counts.

5. MySQL Analysis — Business Questions

The following questions were answered using the healthcare_analytics database and patient_data table.

Q5. How many records belong to each admission type?

Result Grid | Filter Rows:

admission_type	total_records
Elective	18473
Urgent	18391
Emergency	18102

Elective: 18,473; Urgent: 18,391; Emergency: 18,102.

Q6. Which conditions have the highest average billing?

Result Grid | Filter Rows:

medical_condition	average_billing
Obesity	25804.36
Diabetes	25660.48
Asthma	25633.46
Arthritis	25511.78
Hypertension	25503.06
Cancer	25152.32

Obesity is highest at $25,804.36; Cancer is lowest at $25,152.32 in the supplied output.

5. MySQL Analysis — Business Questions

The following questions were answered using the healthcare_analytics database and patient_data table.

Q7. Which admission types have the highest average billing?

Result Grid | Filter Rows:

admission_type	average_billing
Elective	25612.14
Urgent	25514.53
Emergency	25505.33

Elective is highest at $25,612.14, followed by Urgent and Emergency.

Q8. Which hospitals have the highest total recorded billing?

Result Grid | Filter Rows: | Export:

hospital	total_billing	total_records
Sons and Miller	143444.93	7
Kim Inc	145680	6
Cook PLC	103509.11	4
Hernandez Rogers and Vang,	37909.78	1
White-White	28476.64	2

The screenshot shows sample hospital totals. The supplied SQL query groups by hospital but does not sort the output, so this screenshot alone does not establish the true top hospitals.

5. MySQL Analysis — Business Questions

The following questions were answered using the healthcare_analytics database and patient_data table.

Q9. What is the average hospital length of stay?

Result Grid | Filter Rows:

average_length_of_stay_days
15.50

The supplied result reports an average stay of 15.50 days.

Q10. Which hospitals have the longest average stays?

Result Grid | Filter Rows: | Export: | Wra

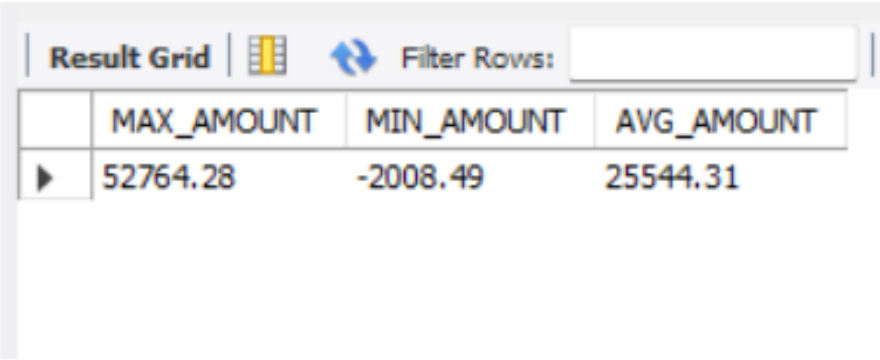
hospital	average_stay_days	total_records
Li Ltd	30.00	1
Simmons and Webster, Mathis	30.00	1
Mullins and Martin Francis,	30.00	1
and Tapia Cole, Casey	30.00	1
and Shaw Sanchez Johnson,	30.00	1

The screenshot shows several hospitals with a 30-day average based on one record each. This should be interpreted cautiously because the displayed groups have very small record counts.

5. MySQL Analysis — Business Questions

The following questions were answered using the healthcare_analytics database and patient_data table.

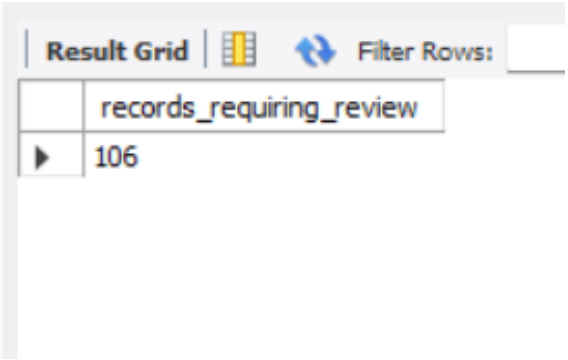
Q11. What are the highest, lowest, and average billing amounts?



	MAX_AMOUNT	MIN_AMOUNT	AVG_AMOUNT
▶	52764.28	-2008.49	25544.31

Maximum: $52,764.28; minimum: -$2,008.49; average: $25,544.31.

Q12. How many records require billing review?



	records_requiring_review
▶	106

The result shows 106 records flagged for review because billing_amount is less than or equal to zero.

5. MySQL Analysis — Business Questions

The following questions were answered using the healthcare_analytics database and patient_data table.

Q13. How many records were admitted each month?

Result Grid | Filter Rows:

admission_month	total_admissions
2019-05	677
2019-06	899
2019-07	951
2019-08	985
2019-09	924

The screenshot displays the first months: May 2019: 677; June: 899; July: 951; August: 985; September: 924.

Q14. Which medications appear most frequently?

Result Grid | Filter Rows:

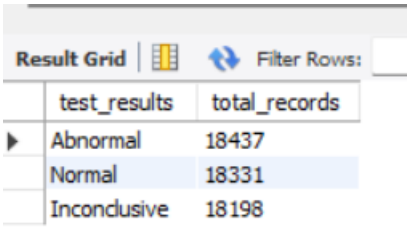
medication	total_records
Lipitor	11038
Ibuprofen	11023
Aspirin	10984
Paracetamol	10965
Penicillin	10956

The screenshot lists Lipitor (11,038), Ibuprofen (11,023), Aspirin (10,984), Paracetamol (10,965), and Penicillin (10,956).

5. MySQL Analysis — Business Questions

The following questions were answered using the healthcare_analytics database and patient_data table.

Q15. What is the distribution of test results?

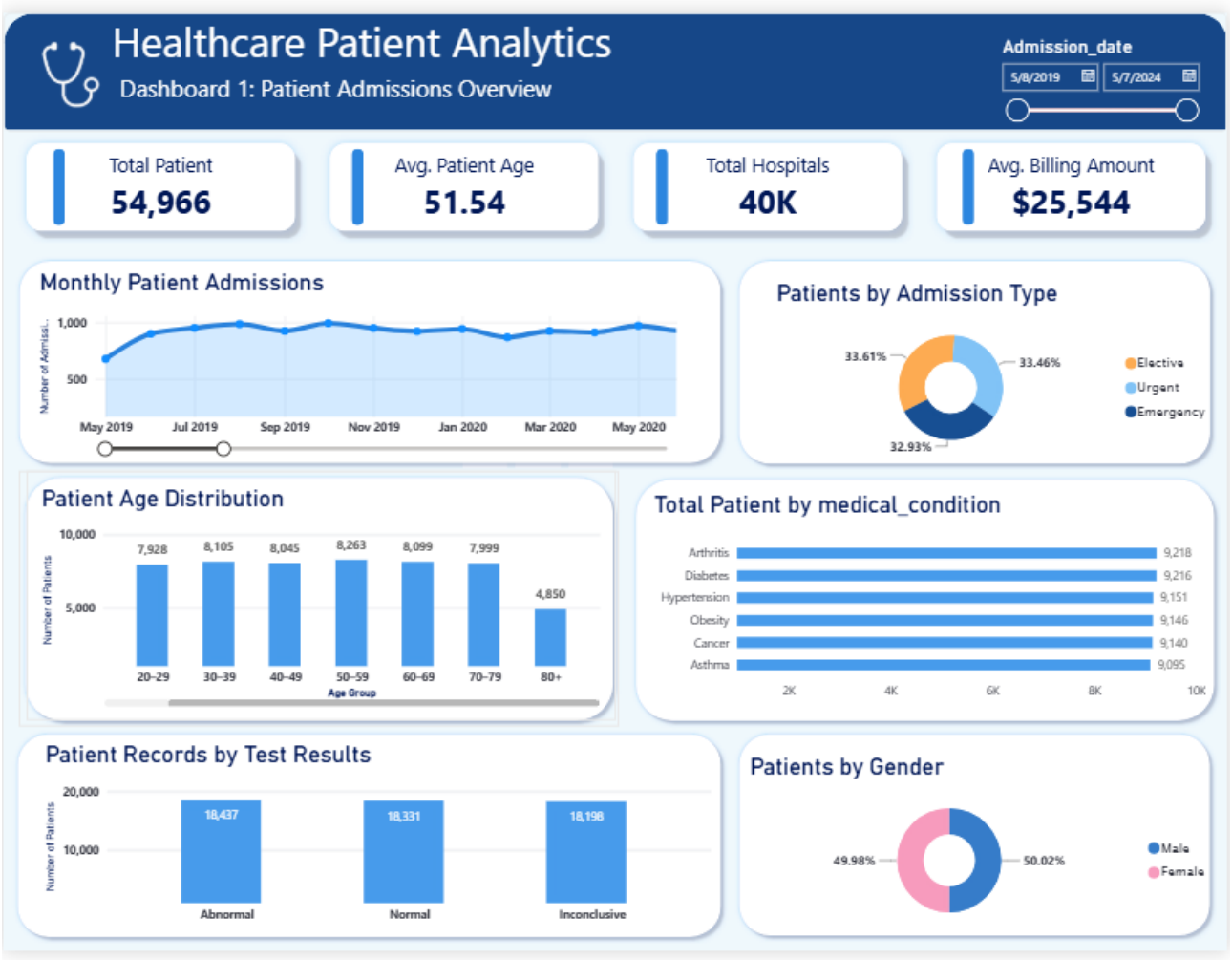


test_results	total_records
Abnormal	18437
Normal	18331
Inconclusive	18198

Abnormal: 18,437; Normal: 18,331; Inconclusive: 18,198.

6. Power BI Dashboard 1 — Patient Admissions Overview

This page summarizes patient-record volume, age, hospital-name count, billing, monthly admissions, admission type, medical condition, test results, and gender.

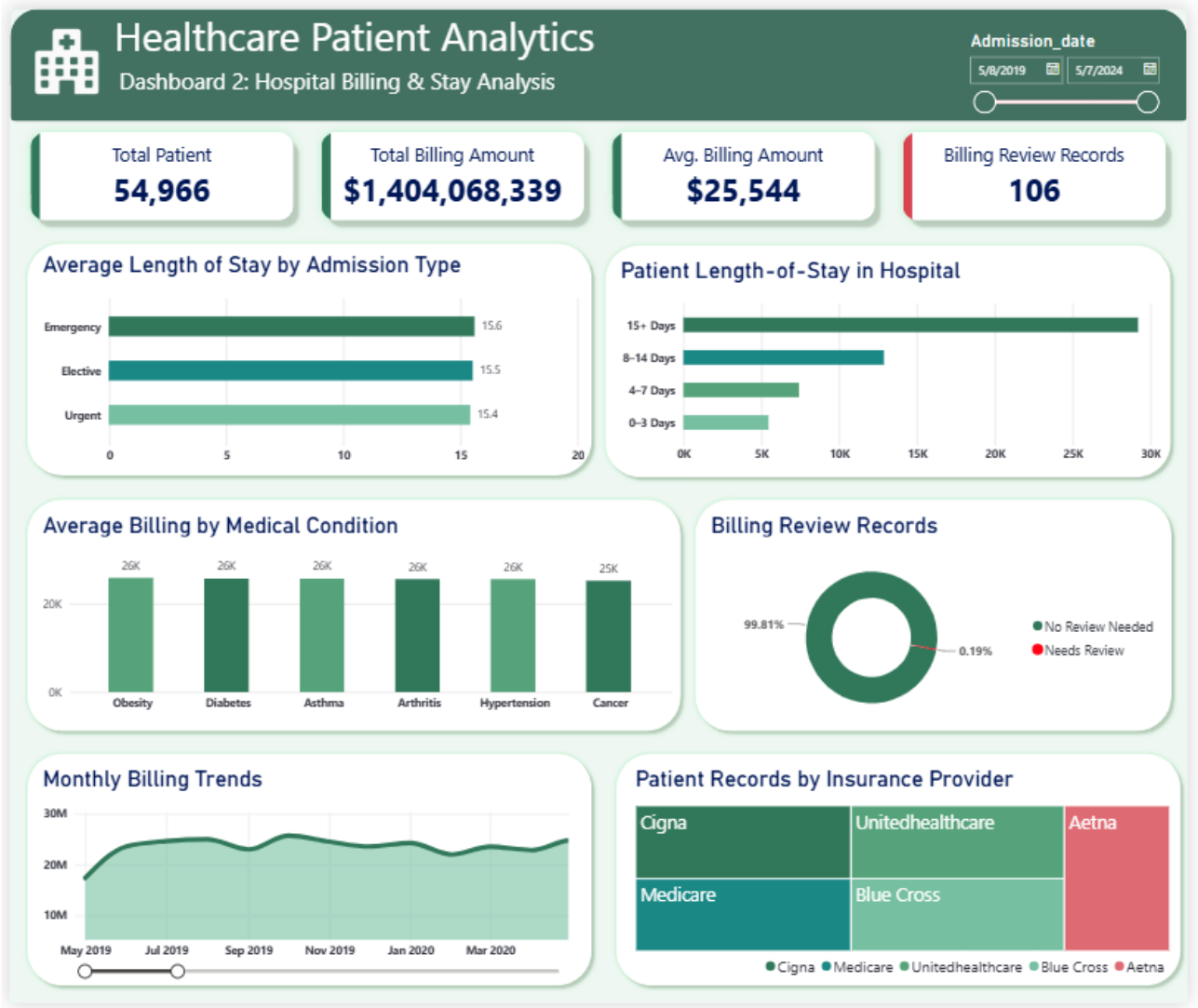


Key elements shown: 54,966 patient records; average age 51.54; average billing amount about $25,544; monthly admission trend; admission-type distribution; age distribution; record counts by medical condition and test result; and gender distribution.

Interpretation note: the card labeled Total Hospitals displays about 40K distinct hospital-name values. Because this is unusually high for a hospital count, it should be described as distinct hospital names in the dataset unless the underlying names have been validated and standardized.

7. Power BI Dashboard 2 — Hospital Billing & Stay Analysis

This page focuses on recorded billing, average length of stay, billing review flags, monthly billing trends, and category comparisons.



The dashboard shows total recorded billing of $1,404,068,339, average billing of about $25,544, average length of stay of 15.50 days, and 106 records flagged for billing review. It also compares average stay by admission type, stay-duration groups, billing by medical condition, monthly billing, and patient records by insurance provider.

Billing-review display: 106 of 54,966 records is approximately 0.19%. The donut makes this small share difficult to compare visually; a horizontal bar or a KPI showing both count and percentage may communicate it more clearly.

8. Business Recommendations

Recommendations below are limited to patterns supported by the supplied notebook outputs, SQL screenshots, and dashboards.

Review flagged billing records

Investigate the 106 records with zero or negative billing amounts (the notebook output identifies 106 negative bills and zero zero-value bills). Confirm whether each is a correction, adjustment, or data-quality issue before changing the values.

Validate hospital-name quality

The dashboard counts about 40K distinct hospital names. Check for inconsistent spelling, punctuation, or formatting before using hospital-level rankings or comparisons.

Use cautious hospital stay comparisons

Some displayed hospital groups have an average stay of 30 days from a single record. Add minimum record-count thresholds before treating these as meaningful comparisons.

Monitor billing patterns by category

Obesity has the highest average recorded billing by medical condition in the notebook output, while Cancer has the lowest. Review underlying distributions and sample sizes before using these differences for planning.

Track admission trends over time

Use the monthly admissions and monthly billing visuals to monitor changes over time. Investigate spikes with additional operational context rather than assuming a cause.

Improve dashboard accessibility

Keep labels readable, use consistent colors, and show both the count and percentage for billing-review records. Consider tooltips for exact billing values when bars appear similar.

9. Tools and Technologies

Python (pandas, NumPy and plotting/EDA workflow), MySQL, and Microsoft Power BI.

Conclusion

This project demonstrates an end-to-end analytics workflow: cleaning a healthcare dataset, exploring patient and billing patterns, answering structured business questions in SQL, and presenting findings in Power BI. The strongest next step is to validate hospital-name quality and review flagged billing records so that future comparisons are based on consistent, interpretable data.